

STATS 418

Color Analysis using Machine Learning

Simplifying style and building a cohesive closet

Presented By: **Julia Fung**



Overview/Motive

Why color analysis?

Problem: While some enjoy the process, getting dressed is unnecessarily difficult for many from celebrities to everyday people.

Motivation: Your clothes should fit you rather than you needing to fit your clothes. Using color analysis, people can filter which clothes suit them and which do not to optimize the process.



Data Collection Approach



Data Source:

ColorMine.ai – Celebrity color analysis directory



Method:

Web scrape each celebrity's information



Progress:

Collected and cleaned each celebrity's job, job category, eye color, hair color, skin tone, contrast, **color season**, and **best/worst colors**

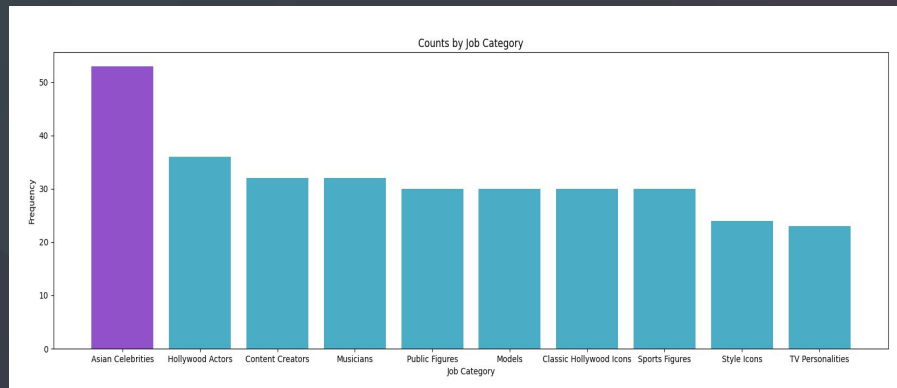
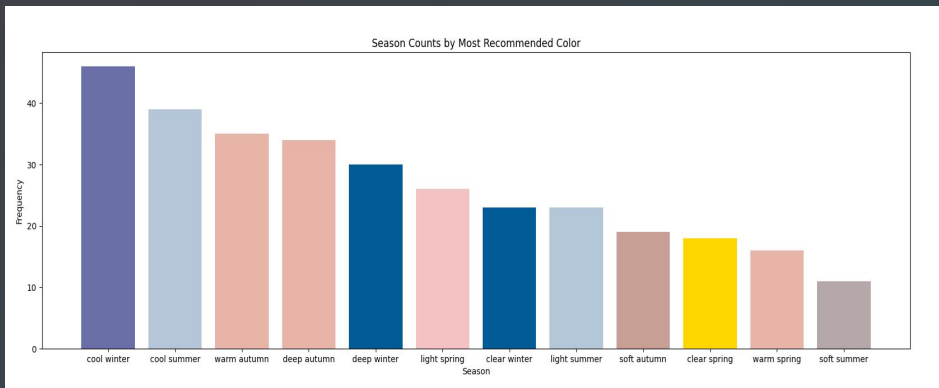
Exploratory Analysis

Key Patterns

- Unbalanced Data
- Overlapping palettes
- Mostly Asian celebrities

Data Processing and Quality

The data was fairly clean with 12 color seasons, but the main cleaning narrows the 137 eye colors to five colors and the 74 hair colors to five as well.



Proposed Model

MLP vs. CNN



Engineered Features:

- Hair color
- Eye color
- Job category



Evaluation Metrics:

- Accuracy
- F1 score
- AUC-ROC
- Loss



Milestones and Timeline:

4/13: choose data	5/2-5/5: EDA
4/20: create environment	5/6-5/17: modeling
4/22-29: get data	5/18-5/26: app
4/30-5/1: clean	5/27-6/2: final presentation

Challenges and Solutions:

- **Small data** → K-fold cv
- **Image inconsistencies** → resize
- **Uneven counts** → change to binary

Remaining Work:

- **Modeling**
- **App**
- **Presentation**

Timeline and Next Steps



Thank You!

Links:

Link to github: https://github.com/jfung07/FungJulia_Stats418_Final

Link to site: <https://colormineai.com/celebrity/>

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Clothing Presentation

Understanding Style, Function, and Expression



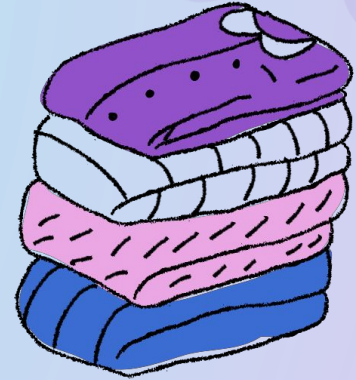
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Presented By: **Avery Davis**

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Introduction to Clothing



More Than Just What We Wear

Clothing is an essential part of human life that serves not only as protection but also as a way to express personality, values, and social identity.



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The Purpose of Clothing



From daily activities to special occasions, clothing plays a key role in providing comfort, safety, and confidence for different situations.

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Protection: Clothing helps protect the body from weather, environment, and external elements.



Comfort: Proper clothing supports movement and enhances physical comfort throughout the day.



Expression: Clothing allows individuals to express mood, personality, and personal style.

Main Functions of Clothing

Why Clothing Matters



Types of Clothing

Different Styles for Different Needs



Casual Wear:

Designed for daily use with a focus on comfort and simplicity



Formal Wear:

Used for professional or special events to create a polished appearance.



Functional Wear:

Created for specific activities such as sports or outdoor purposes.

Clothing Choices

What Influences Our Choices



Occasion:

Different events require different clothing styles and levels of formality.



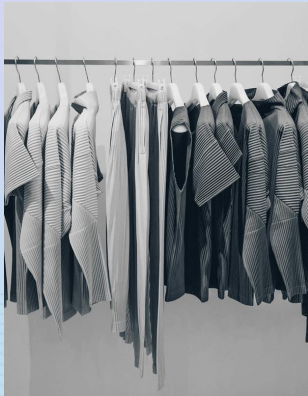
Personal Preference:

Individual taste plays a major role in clothing decisions.



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Clothing and Culture



Clothing often represents cultural background and traditions, showing how fashion connects deeply with history and social norms.

Reflecting Traditions and Values



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Clothing in Modern Life

In modern society, clothing continues to evolve by combining functionality, creativity, and lifestyle needs in everyday fashion.



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The Impact of Clothing

Beyond Appearance

The way people dress can influence first impressions, self-confidence, and how individuals are perceived in social interactions.



Thank you so much



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